

# AI TRAINING – DAY 5 ASSIGNMENT

## Python Loops: Repeat → Control → Break → Continue → Nest

*Learning Guidance + Practical Problems + Student Deliverable*

|                          |                                                    |
|--------------------------|----------------------------------------------------|
| <b>Level</b>             | Beginner → Medium                                  |
| <b>Suggested Time</b>    | 3–4 hours                                          |
| <b>Core Topics</b>       | for, while, range(), break, continue, nested loops |
| <b>Daily Deliverable</b> | Loop Practice Notebook                             |
| <b>Main Focus</b>        | Learn to think using repetition and loop control   |

### Important: What Is the Student Deliverable?

This assignment is your learning guidance. You do not submit a separate file for every activity. Your final submission for Day 5 is ONE notebook called the Loop Practice Notebook. Put your practice, 10 problems, debugging work, real-world challenge and reflection inside that notebook.

Recommended filename: `Day_05_Loop_Practice_YourName.ipynb`

### Learning Goals

- Understand for and while loops.
- Explain loops, range(), break, continue and nested loops in your own words.
- Use range() correctly.
- Understand when to use break and continue.
- Build nested loops.
- Identify common loop errors and improve your solution.
- Solve problems using loops instead of repeating code manually.
- Solve 10 loop problems including prime, palindrome and patterns.

### How to Use This Assignment

1. Read and understand the concept before coding.
2. Predict output before running small examples.
3. Write your own solution before asking AI for help.
4. Run code with different inputs.
5. When code fails, identify and explain the error instead of only replacing the code.
6. Record final work in the Loop Practice Notebook.

### PART A – Learn Before Coding

For every concept, explain it in your own words. Do not copy a definition.

#### 1. for loop

When do I use a for loop?

A for loop is used when the same condition need to check multiple times.

Write 2 simple examples.

Example1:-

For i in range (1,9):

Print(i)----->1 2 3 4 5 6 7 8

Example2:-

for a in "pavithra":

Print(a)----->p a v i t h r a

## 2. while loop

When do I use a while loop instead of a for loop?

A while loop is useful when you don't exact time to reapt the code .

Write 2 simple examples.

Example:-

Num=n

while num!=0:

Num=num//10

While true:

If i==4:

break

## 3. range()

Experiment with:

```
range(5) ----->0,1,2,3,4
range(1, 6) ----->1,2,3,4,5
range(2, 11, 2) ----->2,3,5,6,8,9,11
range(10, 0, -1) ----->10,9,8,7,6,5,4,3,2,1
```

Predict the values before running. Understand:

```
range(start, stop, step)
```

The stop value is normally not included.----->it run until it ends

## PART B – Loop Control

### 1. break

Create a loop that stops when a particular number is reached. Explain what happens when break executes.

For i in range (1,9):

If i == 6;

Break

Here it breaks when i equals to 6.

## 2. continue

Create a loop that prints 1–20 but skips multiples of 3. Explain whether continue stops the entire loop.

For i in the range(1,21):

    If i%3 ==0:

        Continue

Yes the above code skip when the number is divisible by 3.

## 3. Nested Loops

Create:

```
1 2 3
1 2 3
1 2 3
```

Identify the outer loop, inner loop and total inner-loop executions.

For i in range(0,3):

    For j in range(1,4):

        Print(j,end=" ")

    Print()

## PART C – Predict Before You Run

Predict answer then check.

```
for i in range(5):
    print(i)----->0 1 2 3 4

for i in range(1, 6):
    if i == 3:
        break
    print(i)----->1 2

for i in range(1, 6):
    if i == 3:
        continue
    print(i)----->1 2 4 5

i = 1
while i <= 5:
    print(i)
    i += 1----->1 2 3 4 5
```

## PART D – Solve 10 Loop Problemst

### Problem 1 – Multiplication Table

Ask for a number and print its table from 1 to 10. Focus: for + range().

```
number=int(input("enter the number:"))
for i in range(11):
    print(f"{number} x {i} = {i*number}")
```

#### Output:-

enter the number:5

5 x 0 = 0

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50

### Problem 2 – Sum of Numbers

Ask for n and calculate 1 + 2 + ... + n. Focus: accumulator + for.

```
n=int(input("Enter the number :"))
sum=0
for i in range(n+1):
    sum=sum+i
print(f"the sum of whole number from 1 to {n} is {sum}")
```

#### Output:-

Enter the number :37

the sum of whole number from 1 to 37 is 703

### Problem 3 – Count Even and Odd Numbers

For 1 to n, count even and odd numbers. Focus: for + if.

```
num=int(input("Enter the number:"))
even_count=0
```

```

odd_count=0
for i in range(1,num+1):
    if i%2 == 0:
        even_count +=1
    else:
        odd_count +=1
print(f"the number of even numbers in range of 1 to {num} are {even_count}")
print(f"the number of odd numbers in range of 1 to {num} are {odd_count}")

```

#### Output:-

Enter the number:231

the number of even numbers in range of 1 to 231 are 115

the number of odd numbers in range of 1 to 231 are 116

#### Problem 4 – Prime Number Checker

Determine whether a number is prime. First explain what makes a number prime.

The number which is divisible by only the 1 and itself it know as prime number.

```

number1=int(input("enter the number which you want to check (prime /not )"))
for i in range (2,number1):
    if number1%i == 0:
        print(f"{number1} not a prime number")
        break
else:
    print(f"{number1} is a prime number.")

```

#### Output:-

enter the number which you want to check (prime /not )9864

9864 not a prime number

#### Problem 5 – Palindrome Number

Check whether a number such as 121 is a palindrome. Try without a ready-made palindrome function.

```

number2=int(input("enter the number: "))

```

```

num1=number2
r_number2=0
while num1!=0:
    digit=num1%10
    r_number2=(r_number2*10)+digit
    num1=int(num1//10)
if number2==r_number2:
    print(f"{number2} is a palindrome number")
else:
    print(f"{number2} not a palindrome number.")

```

#### Output:-

enter the number: 767

767 is a palindrome number

enter the number: 5678

5678 not a palindrome number.

#### Problem 6 – Find Factors

Print all factors of a number. Example: 12 → 1, 2, 3, 4, 6, 12.

```

number3=int(input("enter the number :"))
print("the factors are :")
for i in range (1,number3+1):
    if number3%i== 0:
        print(i)

```

#### Output:-

enter the number :87

the factors are :

1

3

29

87

### Problem 7 – break Challenge

Search 1–100 and stop at the first number divisible by both 7 and 9. Use break.

```
for i in range(1,101):  
    if i%7 == 0 and i%9 == 0:  
        print(f"the frist number for 1 to 100 which is divisible my both 7 and 9 is {i}")  
        break
```

Output:-

the frist number for 1 to 100 which is divisible my both 7 and9 is 63

### Problem 8 – continue Challenge

Print 1–50 but skip numbers divisible by 3. Use continue.

```
for i in range(1,51):  
    if i%3==0:  
        continue  
    print(i)
```

Output:-

1 2 4 5 7 8 10 11 13 14 16 17 19 20 22 23 25 26 28 29 31 32 34 35 37 38 40 41 43 44 46 47 49 50

### Problem 9 – Pattern Printing

Print \*, \*\*, \*\*\*, \*\*\*\*, \*\*\*\*\* and the reverse. Bonus: centered pyramid.

```
for i in range(1,6):  
    print(""*i)  
print("-----")  
for i in range(1,6):  
    print(""*(6-i))  
print("-----")  
for i in range(1,6):  
    for j in range(5-i):  
        print(" ",end="")  
    for j in range(2*i-1):  
        print("*",end="")  
    print()
```

Output:-

```

*

**

***

****

*****

-----

*****

****

***

**

*

-----

    *

    **

    ***

    ****

    *****

    ******

    *******

```

### Problem 10 – Number Pattern

Print 1, 12, 123, 1234, 12345 and then 1, 22, 333, 4444, 55555. Use nested loops.

```

for i in range(1,6):
    for j in range(1,i+1):
        print(j,end="")
    print()
print("-----")
for i in range(1,6):
    for j in range(1,i+1):
        print(i,end="")
    print()

```

Output:-

```

1
12

```

123

1234

12345

-----

1

22

333

4444

55555

## PART E – Debugging Lab

For every bug: identify the problem, explain why it happens, fix it and test it.

```
Bug 1
for i in range(5)
    print(i)
```

What is missing?-----> ":" missing

```
for i in range(5):
    print(i)
```

```
Bug 2
i = 1
while i <= 5:
    print(i)
```

Why can this become an infinite loop? Fix it.----->because there is no change without increment.

```
i = 1
while i <= 5:
    print(i)

    i+1
```

```
Bug 3
for i in range(1, 10):
    if i == 5:
        break
    print(i)
    print('Loop completed')
```

Understand where the loop actually stops.----->it break when i equals to 5output is 1 2 3 4

```
for i in range(1, 10):
    if i == 5:
```

```

        print('Loop completed')
        break
    print(i)

```

Bug 4

```

for i in range(1, 10):
    if i % 2 == 0:
        continue
    print(i)

```

Why will print(i) never execute for even numbers?----->because the continue skip the loop when it is an even number.

```

for i in range(1, 10):
    if i % 2 == 0:
        print(i)

```

Bug 5

```

for i in range(3):
    for j in range(3):
        print('*')

```

Predict how many times \* prints, then verify.----->9 times the \* prints.

## PART F – Real-World Challenge

Choose ONE: ATM transaction loop; Login system with 3 attempts; Shopping cart total calculator; Student marks analyzer; Number guessing game.

- Use a loop.
- Use at least one if.
- Use at least one break OR continue.
- Take user input.
- Display proper output.
- Test at least 3 different cases.

```

ATM_pin=123

i=1

while i<=3:

    pin=int(input("Enter your pin :"))

    if pin==ATM_pin:

        print("Access Granted")

        break

    if i == 3 and pin != ATM_pin:

        print("Card Blocked")

        break

```

## PART G – Explain in Your Own Words

7. What is a loop?  
It repeats the statement block when it is correct.
8. What is the difference between for and while?  
For is used to when you know exact time of loops and while is used for you don't know exact times.
9. What does range() do?  
Range does where to start where to stop and how many jumps to take.
10. What happens when break executes?  
Break is used to break the loop.
11. What happens when continue executes?  
It skips the particule time.
12. What is a nested loop?  
Loop in another loop.
13. When can a while loop become infinite?  
The condition always true.
14. Why is indentation important inside loops?  
It tells which the block belongs to which loop or statement condition
15. What was the hardest problem today?  
Patterns printing ..
16. What mistake did you make and how did you fix it?  
I make mistake while writing patterns that for loops in another loop.while compare my output and make the change then I clarify that.

## PART H – Self Check

| Skill            | Comfortable | Need Practice                |
|------------------|-------------|------------------------------|
| for loop         | yes         | <input type="checkbox"/>     |
| while loop       | yes         | <input type="checkbox"/>     |
| range()          | yes         | <input type="checkbox"/>     |
| break            | yes         | <input type="checkbox"/>     |
| continue         | yes         | <input type="checkbox"/>     |
| Nested loops     | yes         | <input type="checkbox"/>     |
| Prime number     | yes         | <input type="checkbox"/>     |
| Palindrome       | yes         | <input type="checkbox"/>     |
| Pattern printing | yes         | <input type="checkbox"/> yes |
| Debugging loops  | yes         | <input type="checkbox"/> yes |

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## Submission Checklist

- ☐ Concept practice completed. ✓
- ☐ All 10 loop problems completed. ✓
- ☐ Prime number completed. ✓
- ☐ Palindrome completed. ✓
- ☐ Pattern problems completed. ✓
- ☐ Debugging completed. ✓
- ☐ Real-world challenge completed. ✓
- ☐ Reflection completed. ✓

- ☐ Test cases included.✓

## **DAY 5 PRINCIPLE**

*Don't just repeat code. Learn to control repetition.*